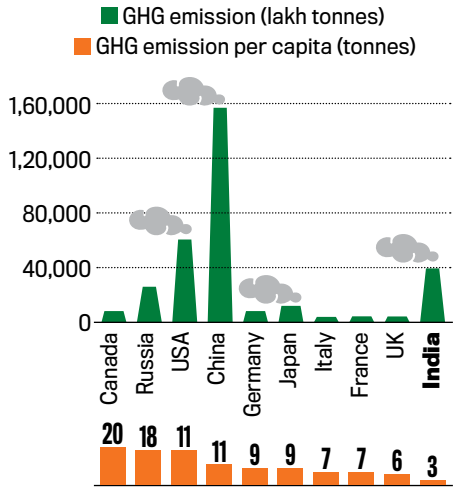
THE NEW POWER GAME

India must reduce its dependence on coal and harness alternative sources for power generation

MARCHING AHEAD

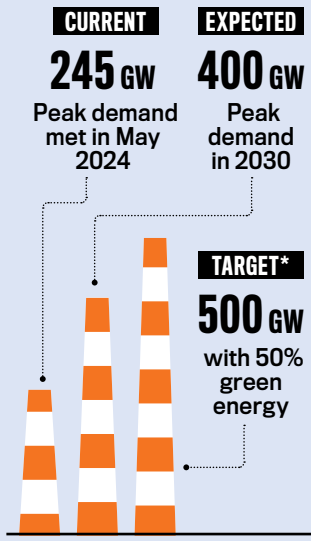
India is among world's top 10 polluters but low on per capita emissions



Source: Emissions Database for Global Atmospheric Research (EDGAR)

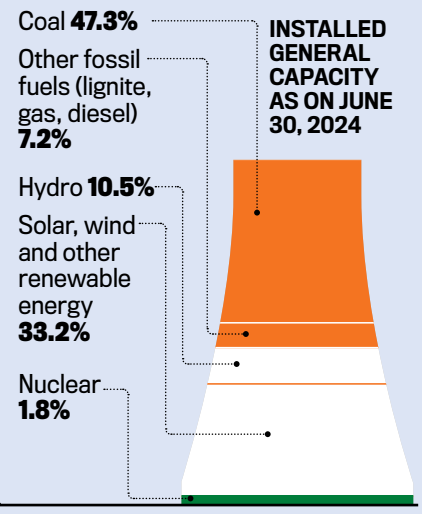
POWER PROJECTIONS

(Current and expected demand for power)



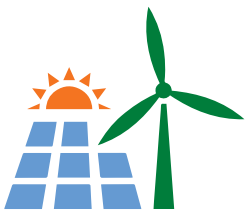
THE ENERGY MIX

Fossil fuels remain the dominant source of energy in the country



*Source: NITI Aayog and Central Electricity Authority; *by 2030*

to diversify their supply chains. India can tap into the opportunity and formalise partnerships with developed countries to supply these products. There is, however, a need to focus on sections of the value chains where it is, or can be, competitive at a global scale without government support, which will ensure the long-term sustainability of the sector." Exports, adds Rahul Tongia, senior fellow with the Centre for Social and Economic Progress in New Delhi, should be driven by strategy, not revenue targets. "The domestic market itself has a voracious appetite, and we should aim to prioritise that demand. Export premiums are also important and the two should ideally not be mutually exclusive. India should aim at emerging as a player exporting full-fledged services critical for uptake in the entire supply chain, not just commoditise the cheapest products. Countries may have a China-plus-one strategy, but that cannot be sustained just on any cost advantage we might offer, which itself is tough. China has slack capacity, so it's hard to beat them on cost."



TARGET 2030

- Reduce economy's carbon intensity by 45% over 2005 levels
- Meet 50% of energy requirements from renewables
- Build 500 GW non-fossil energy capacity by 2030
- Reduce total projected carbon emissions by 1 billion tonnes by 2030
- Aim for 5 MT annual production capacity of Green Hydrogen, and 125 GW of renewable energy
- Develop 50 solar parks with aggregate 37.5 GW capacity
- Keep offshore target of 30 GW for wind energy
- Target 27 GW from PHS (pumped hydro storage) and 47 GW from BESS (Battery Energy Storage System) by 2032

COAL, THE ELEPHANT IN THE ROOM

At the climate change summits, much to the displeasure of developed nations, India has steadfastly asserted that it will not abandon coal but will minimise it, since its lower income sections cannot bear the burden of the steep additional cost. Since coal accounts for nearly 70 per cent of India's power generation, the thrust is on increasing the efficiency of thermal power plants and deploying the latest technology to reduce emissions in the supply chain, from coal mining to transportation to storage. The government has also launched clean coal initiatives like the Coal Gasification Mission, which aims to gasify 100 million tonnes of coal by 2030. This will mean the adoption of advanced ultra-supercritical (AUSC) technologies. AUSC plants operate at much higher temperatures and pressures than conventional coal-fired power plants, improv-